

Predicting Carboxymethyl Cellulase assay (CMCase) production using Artificial Neural Network and explicit feature selection approach

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Abstract—This paper presents a method for predicting carboxymethyl cellulase (CMCase) production using artificial neural networks (ANNs) and an explicit feature selection approach. A dataset of CMCase production experiments was collected, and an explicit feature selection approach was applied to select the most relevant features for CMCase production prediction. The ANN model was trained using both the selected features and all available features of the CMCase production data. The results showed that the explicit feature selection approach improved the performance of the ANN model in terms of prediction accuracy compared to using all the features available in the dataset. The main effect analysis (MEA) was found to be the best method for selecting the explicit features for predicting CMCase production. The ANN model trained using the MEA identified features, achieved 96.3% R^2 score and a MAE of 0.057 and a MSE of 0.035. The proposed method is an effective approach for predicting CMCase production and can be used to optimize CMCase production and reduce costs in various industries.

Index Terms—CMCase, enzyme production prediction, Artificial Neural Network, Random forest regressor feature selection, main effect analysis feature selection

I. INTRODUCTION

Carboxymethyl cellulase (CMCase) is an essential enzyme in the cellulose hydrolysis process [1], which has a wide range of industrial applications such as biofuel production, papermaking, and textile manufacturing. The efficient and cost-effective production of CMCase is crucial for the success of these industries. The production of CMCase is influenced by various factors, such as substrate concentration, pH, temperature, and agitation speed, among others [2]. Carboxymethyl cellulase (CMCase) production prediction is important for several reasons. The first one is optimizing production. By predicting the production of CMCase, it is possible to design optimal fermentation conditions, which can lead to improved production efficiency and cost-effectiveness. This can help

lower the production costs and increase the overall yield of CMCase.

Quality control is another reason. CMCase production prediction can be used to monitor and control the quality of the produced enzyme, ensuring that it meets the required specifications. This can help to maintain consistency in the production of CMCase. CMCase production prediction can also be used to develop new processes for the production of CMCase, such as identifying new substrate sources or developing new fermentation conditions. Environmental sustainability is another major reason for the CMCase production prediction. CMCase is widely used in various industries, such as biofuel production, papermaking, and textile manufacturing. By predicting the production of CMCase, it is possible to optimize the use of resources and reduce waste, which can help improve environmental sustainability [3] and reducing the costs of using the substrate and energy required for the fermentation process.

In this study, we proposed a method for predicting CMCase production using artificial neural networks (ANNs) and an explicit feature selection approach using random forest regressor and main effect analysis methods. The proposed method has significance in optimizing the production of CMCase and reducing costs in various industries. It enables forecasting of CMCase yield and sets optimal fermentation conditions for efficient and cost-effective production. Artificial neural networks (ANNs) are particularly suitable for this purpose, as they can effectively process high-dimensional data and model intricate relationships between inputs and outputs. A feature selection technique is also crucial, as it enhances the performance of the ANN model by identifying the most vital features for CMCase production prediction.

The paper is organized as follows: Section II describes

the related works on artificial neural networks in predicting enzyme production. Section III presents the methodology implemented in the study. Section IV analyzes the result of experiments conducted on the CMCase dataset using ANN and explicit feature selection methods. Finally, section V concludes the findings of the study.

II. RELATED WORK

There is a limited amount of research on the use of Artificial Neural Networks (ANN) [4] in predicting the production of CMCase. Response Surface Methodology (RSM) [5], Adaptive Neuro-Fuzzy Inference Systems (ANFIS) [6], and Genetic Algorithms (GA) [7] for the prediction of CMCase production are being used on the other hand. While ANN has been widely studied in various fields, its application in CMCase production prediction is not well researched. Similarly, RSM, ANFIS, and GA have also been used in a limited number of studies in this specific area.

The work by S. Ratnakomala et al [8], discusses the use of Taguchi experimental design and Response Surface Methodology (RSM) to optimize the fermentation medium for increasing carboxymethyl cellulase (CMCase) production by marine actinomycetes *Streptomyces* sp. Bse 7-9. The Taguchi experimental design was used first to identify the effect of factors such as bagasse on other factors, then Central Composite Design was used to determine the interaction between factors to find the optimal concentration of ingredients. However, the authors worked only on optimizing the output based on the ingredient selection; no prediction model was provided in the study.

Another work by Nisar et al [9], describes the use of an experimental design to optimize nutritional parameters for carboxymethyl cellulase production. The study used a central composite design with 8 axial points, 16 factorial points, and 6 center points to optimize and predict the effect of four independent variables, including lactose, peptone, ammonium sulphate, and Tween 80. The study used ANOVA to test the significance of the four independent variables and their possible interactions with a 95% confidence level.

The study by Nelofer et al [10] aimed to produce carboxymethyl cellulase by submerged fermentation and optimize saccharification through Box-Behnken design. Three variables (pH, incubation temperature, and substrate concentration) were used to optimize the saccharification of wheat straw (83% cellulose) using Box Behnken design and artificial neural network. The results showed that the proposed model had a significant and quadratic effect on these parameters that significantly affected sugar production. The ANN model proposed in this work achieved 95.6% R^2 score showing the potential of the ANN.

Despite the lack of literature in this area, the potential of an artificial neural network for improving CMCase production prediction is significant. Thus, there is a need for more research in this field to fully explore the capabilities of ANNs with explicit feature selection techniques.

III. METHODOLOGY

The methodology for this study includes several steps to predict CMCase production using ANN. First, lab scale experimental data is collected from various sources relevant to CMCase production. Next, feature selection is performed using both Random Forest Regression (RFR) [11] and Mutual Information-based Feature Selection (MEA) [12] to identify the most informative features for the prediction task. These features are then used in the proposed ANN model. In addition to using all available features, the ANN model has also been implemented using the features identified through RFR and MEA feature selection to compare the performance of the model with and without feature selection, and its performance has been evaluated using appropriate metrics, namely, mean absolute error (MAE), mean squared error (MSE), and R^2 score. Fig. 1 shows the overview of the methodology employed in this work.

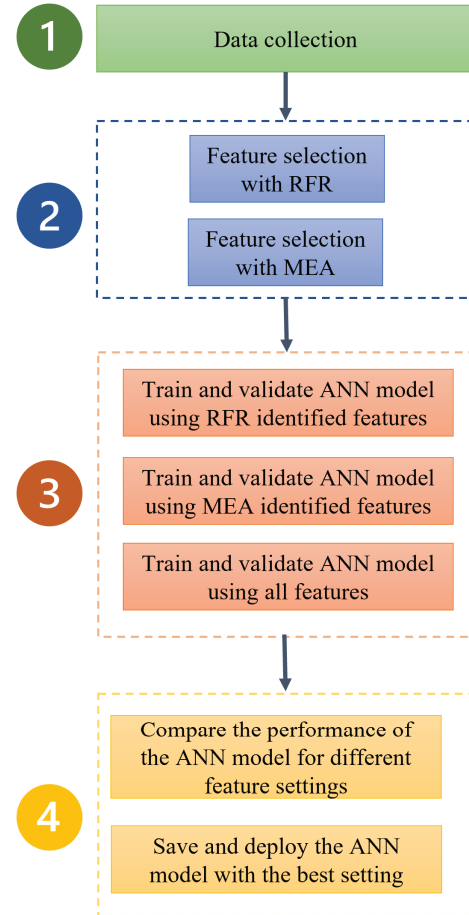


Fig. 1. Overview of the methodology employed in this work

A. Data collection

To predict CMCase production, we collected a dataset of CMCase production experiments. The dataset included various factors such as total sedimented solids of POME, cassava powder, wheat powder, table sugar, cellulose powder,

peptone, (NH₄)₂SO₄, KH₂PO₄, TWEEN 80, MnSO₄·H₂O, MgSO₄·7H₂O and the corresponding CMCase production values from lab-scale experiments. After that, a synthetic dataset is generated using gretel-synthetics [13]. The synthetic dataset contains information on CMCase production from 39 lab-scale experiments, with a total of 5000 data points. The detailed report on the generation of this synthetic data can be found at the following url: https://ping543f.github.io/gretel_report/index.html. Table I illustrates the structure of the dataset, which comprises 12 columns. The first 11 columns represent features, while the last column represents the target. Each column is labeled with an English letter, with the exception of the target column.

TABLE I
DATASET OVERVIEW

Column Number	Column Name	Feature Name
1	A	TSS of POME
2	B	Cassava Powder
3	C	Wheat Powder
4	D	Table Sugar
5	E	Cellulose Powder
6	F	Peptone
7	G	(NH ₄) ₂ SO ₄
8	H	KH ₂ PO ₄
9	I	TWEEN 80
10	J	MnSO ₄ ·H ₂ O
11	K	MgSO ₄ ·7H ₂ O
12	CMCase	CMCase

B. Feature selection

We applied an explicit feature selection approach to select the most relevant features for CMCase production prediction. We have used the RFR and MEA methods to select the features. The top five features have been selected using both methods based on their feature importance score.

In RFR, the feature importance score is calculated by averaging the impurity decrease of each feature over all decision trees in the forest. The impurity decrease is the difference between the impurity of the node before and after the split, weighted by the number of samples in each node. The feature importance score is normalized so that the sum of all feature importances is equal to one.

Let F_i be the i^{th} feature in the dataset, and let $I(F_i)$ be its importance score. Random Forest Regression calculates $I(F_i)$ by computing the mean decrease impurity of F_i across all decision trees in the forest:

$$I(F_i) = \frac{1}{N_{trees}} \sum_{j=1}^{N_{trees}} \sum_{t \in T_j} p_t \Delta i(t, F_i)$$

Where N_{trees} is the number of decision trees in the forest. T_j is the set of nodes in the j^{th} decision tree. p_t is the proportion of samples in node t . $\Delta i(t, F_i)$ is the impurity decrease of node t when splitting on feature F_i . The impurity decrease of a node is computed as:

$$\Delta i(t, F_i) = i(t) - \frac{N_{left}}{N_t} i(t_{left}) - \frac{N_{right}}{N_t} i(t_{right})$$

Where $i(t)$ is the impurity of node t . t_{left} and t_{right} are the left and right child nodes of t , respectively. N_t , N_{left} , and N_{right} are the number of samples in nodes t , t_{left} , and t_{right} , respectively.

In MEA, feature importance is calculated by examining the coefficients of the linear regression model fitted to each feature. Features with larger absolute coefficients are considered more important because they have a larger effect on the response variable.

Let F_i be the i^{th} feature in the dataset, and let b_i be the coefficient of F_i in the linear regression model. The feature importance score $I(F_i)$ can be calculated as:

$$I(F_i) = |b_i|$$

Where $|b_i|$ is the absolute value of the coefficient of F_i .

After selecting the features using the RFR and MEA, they are used to train and validate the proposed ANN model.

C. ANN model development

We trained an ANN model using the selected features and the CMCase production data. The ANN model consisted of an input layer, three hidden layers, and an output layer. The number of neurons in the hidden layers has been determined using a grid search cross-validation approach, and the model has been trained using TensorFlow [14] and Keras [15] libraries. The model has been optimized to avoid overfitting by using early stopping callback and evaluated on a 70/30 train/test split. The architecture of the ANN model is shown in Fig. 2. The detail explanation of the model is discussed below.

- **Input layer:** The ANN model has an input layer that takes the specified features as input. These features could be any measurable or observable properties that are relevant to predicting CMCase production. The number of input nodes in the input layer would be equal to the number of input features.
- **Hidden layers:** The ANN model has three hidden layers that are configured with 256, 512, and 256 units respectively. These layers employ Rectified Linear Unit (RELU) activation, which is a commonly used activation function in neural networks. The number of neurons in each hidden layer is determined through a grid search cross-validation approach, which involves trying different numbers of neurons and selecting the configuration that results in the best performance.
- **Output layer:** The output layer of the ANN model provides the predicted CMCase production. This layer has a single node since we are trying to predict a single continuous output variable.
- **Activation function:** The activation function in the hidden layers is RELU, which is a non-linear activation function that helps the model to learn complex relationships between the input features and the output variable.

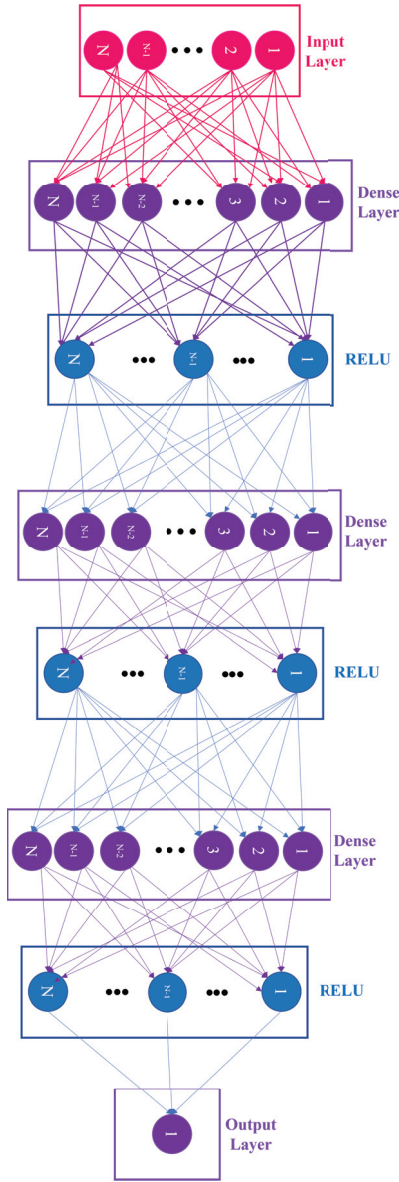


Fig. 2. Architecture of the proposed ANN model

- **Libraries used:** The model has been implemented and trained using TensorFlow, which is an open-source machine learning library developed by Google. The Keras library has also been used to build the model, which is a high-level neural networks API that is built on top of TensorFlow.
- **Training approach:** The optimal number of neurons and batch size in the hidden layers have been determined using a grid search cross-validation approach. The model training epochs are optimized with the early stopping callback setting, which is a technique that helps to avoid overfitting the model to the training data. The training and validation data have been split into a 70% training set and a 30% validation set, which is a common approach to evaluate the performance of a machine learning model.

In summary, the proposed method includes collecting a dataset of CMCase production experiments, applying an explicit feature selection approach to select the most relevant features, and training an ANN model using the selected features and all features of the CMCase production data. The ANN model has been trained using Tensorflow and Keras. Experimental code has been written in the Python programming language [16] and as the model training environment, google colaboratory [17] has been utilized. The code and data of the experiment can be found in the following github repository: <https://github.com/ping543f/cmcase-conf>.

IV. RESULT AND DISCUSSION

During the feature selection process, both RFR and MEA were used to determine the importance of each feature. The top five features with the highest importance scores were then used to train the ANN model. Random Forest Regression (RFR) has identified TSS of POME, Peptone, KH₂PO₄, (NH₄)₂SO₄, and table sugar as the most important features. Meanwhile, the Main Effect Analysis (MEA) method has identified TSS of POME, cellulose powder, KH₂PO₄, TWEEN 80, and MgSO₄·7H₂O as the top five features. Following feature selection, they are used in the ANN model alongside all other features.

The features identified by the RFR and MEA methods are summarized in Table II. They are listed in order of importance, with the most important feature appearing first and the remaining features listed in descending order of importance.

TABLE II
FEATURES IDENTIFIED BY THE RFR AND MEA FEATURE SELECTION METHODS

Feature Importance	RFR	MEA
1	A: TSS of POME	A: TSS of POME
2	F: Peptone	E: Cellulose Powder
3	H: KH ₂ PO ₄	H: KH ₂ PO ₄
4	G: (NH ₄) ₂ SO ₄	I: TWEEN 80
5	D: Table Sugar	K: MgSO ₄ ·7H ₂ O

Once the features were identified, they were utilized to train an ANN model. Three different sets of features were used to train the model. The model was initially trained with all the features present in the dataset. During this phase, the ANN model achieved an R^2 score of 96.2%, an MAE of 0.059, and a MSE score of 0.037. Then the ANN model was trained using the features identified by RFR and achieved an R^2 score of 89.3%, a MAE of 0.157 and a MSE of 0.105. Next, the ANN model was trained using the features identified by MEA and achieved an R^2 score of 96.3%, a MAE of 0.057 and a MSE of 0.035. The results of the experiment are shown in Table III. The comparison of R^2 scores among different feature combinations obtained during the experiment is presented in Fig.3, and the comparison of MAE and MSE scores of the model for different feature combinations is illustrated in Fig.4.

From the experimental results, it is clear that the ANN model performed better with the features identified by the

TABLE III
SCORES ACHIEVED FOR DIFFERENT EVALUATION METRICS FOR
DIFFERENT COMBINATIONS OF FEATURES

Metric	All features	RFR features	MEA features
MAE	0.059611	0.157058	0.0570889
MSE	0.037112	0.105289	0.0356436
R^2	0.962446	0.893458	0.9639323

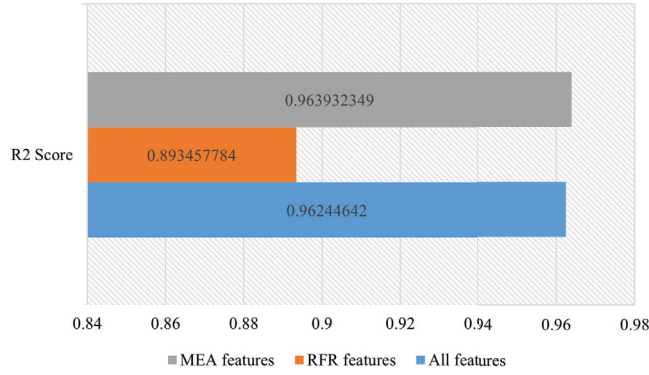


Fig. 3. Comparison of R^2 scores for different combination of features



Fig. 4. Comparison of MAE and MSE scores for different combination of features

MEA feature selection method. This is likely because MEA assumes linear relationships between the predictors and the response variable, and if the relationships are indeed linear, it performs better than RFR. Additionally, MEA is better at handling correlated features by identifying the most important feature among correlated features, whereas RFR might select correlated features. The lowest MAE and MSE scores among all implemented feature sets were also achieved with MEA-identified features.

The performance of the ANN model during training and validation for all experimental combinations in terms of loss is illustrated in Fig. 5. As seen in the figure, there is no indication of overfitting or underfitting, despite the variations in the number of features used to train and validate the model, which also affected the number of epochs required for convergence.

Our findings indicate that the utilization of an explicit fea-

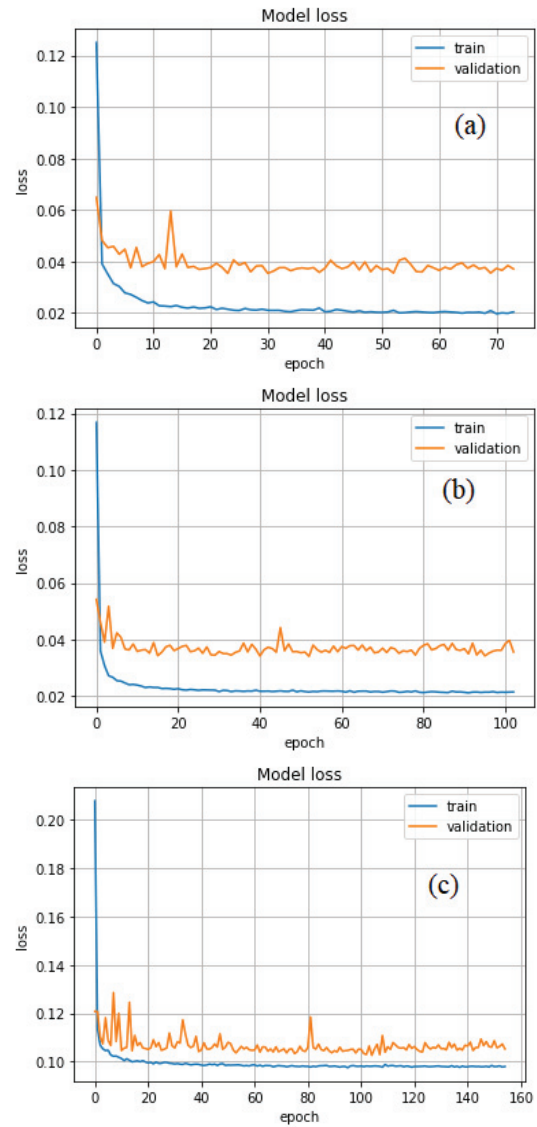


Fig. 5. Training and validation curves for model loss metric of the proposed ANN model for (a) all features (b) MEA identified features and (c) RFR identified features.

ture selection approach enhances the predictive accuracy of the ANN model. The ANN model exhibited a strong correlation coefficient (R) of 0.963 and a small mean squared error (MSE) of 0.035 when compared to actual CMCase production values using the features from MEA. These results suggest that the proposed method is a viable approach for predicting CMCase production.

V. CONCLUSION

In conclusion, we have proposed a method for predicting CMCase production using ANNs and an explicit feature selection approach. Our results showed that the proposed method is an effective approach for predicting CMCase production, with a high correlation coefficient and a low MSE. This approach can be used to optimize CMCase production and

reduce costs in various industries. Further research can be done to improve the prediction accuracy by using other machine learning models or incorporating more data.

ACKNOWLEDGMENT

The work reported in this paper is funded by Trans-Disciplinary Research Grant Scheme from the Ministry of Higher Education Malaysia titled: An Artificial Neural Network Sine Cosine Algorithm-based Hybrid Prediction Model for the Production of Cellulose Nanocrystals from Oil Palm Empty Fruit Bunch TRGS/1/2018/UMP/01/1/4 (RDU191801-4)

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